STATEMENT OF PURPOSE

Hyungjoo Chae
https://hyungjoo-homepage.netlify.app

Consider a developer asking an agent to create a simple personal website, or a user requesting it to order headphones on Amazon. I still vividly remember the first time I saw a digital agent in action – how fascinatingly it independently browses the internet, makes decisions, and completes a task. Whether it was a developer asking the agent to create a simple personal website or an online shopper instructing it to order headphones on Amazon, I was captivated by the agents' ability to explore and interact with the digital world autonomously.

This initial awe gradually deepened as I began to see their broader potential – not just as passive tools, but as transformative assistants capable of revolutionizing how humans interact with computers. Over time, my fascination has evolved into the recognition of digital agents' capacity to grow beyond basic task execution and become intellectual partners, adept at understanding and performing complex high-level instructions comparable to humans, so that humans can focus on more meaningful tasks. For instance, a marketing analyst currently at Apple needs to compile a report on customer feedback for a new iPhone and has to tediously browse through various social media platforms, such as X and Reddit, to generate insightful figures for her report. I envision a future society where digital agents could automate such time-consuming tasks, freeing humans to concentrate on intellectual pursuits such as strategic thinking and creative innovation.

However, my experience with these agents revealed a significant gap between this vision and the current reality. While they excel at simple tasks like finding directions from Georgia Tech to Starbucks, they often struggle with more complex tasks that require long-term planning and reliable execution. Upon the exploration driven both by my genuine curiosity and ongoing research on digital agents, I have come to focus on two main research interests: (1) improving digital agents to self-improve through code generation and (2) building universal digital agents that can plan for long-horizontal tasks.

1 Code as a Magic Wand for Digital Agents

Throughout my research journey, I have come to view code as more than just a programming tool — it is a powerful bridge between human intention and computer execution [5]. My fascination with code stems from its unique ability to break down complex problems into manageable steps, systematically verify our progress, and deliver consistent results [3, 7]. Additionally, its reusable nature is particularly effective in reducing time spent on repetitive tasks, making it an invaluable tool for automation and efficiency.

Empowering Code Agents through Self-debugging. Working with code agents revealed a critical challenge: while they can generate plausible-looking code, they often struggle when it comes to producing perfectly functioning solutions. Through this observation, I discovered that enabling agents to debug their own code could be the most promising path forward. Inspired by the recent success of RLHF and recognizing the limitations of supervised fine-tuning, I reformulated this challenge as a reinforcement learning (RL) problem with a novel reward function based on test case pass rates. In [6], we demonstrated how this method significantly enhanced agents' ability to identify and fix their own coding errors, even with limited training data. This breakthrough particularly excited me as it showed how agents could learn to improve their own code, much like human programmers.

Verifying Actions with Code. During my research with digital agents, I discovered a fundamental limitation that became increasingly apparent. While they often generate correct high-level plans, they frequently stumble when grounding these plans into precise actions. This challenge resonated with my experience in code generation, where we successfully addressed similar correctness issues through self-debugging approach. This parallel inspired me to explore whether we could apply code-based verification to improve agents' grounding capabilities. Our recent success with [1], where we verified LLMs' reasoning on biomedical documents using knowledge base information, further convinced me of this direction's potential. I am particularly excited about extending this verification approach

to digital agents, enabling them to verify and refine their grounding process through code-based feedback loops before executing actions.

Leveraging Code Generation within GUI Environment. A crucial advantage of programming I have discovered is that it unlocks access to a vast ecosystem of powerful tools written in codes that are impossible to access through GUI interfaces alone. These insights have inspired an ambitious future direction I plan to explore during my doctoral studies — building a benchmark that evaluates how agents can effectively utilize both programming tools, such as code editors (e.g., VSCode) and terminals, alongside GUI interactions for solving computer tasks.

2 Learning to Solve Increasingly Complex Computer Tasks

Planning with World Models. During the investigation of the current web agents, I found that the current web agents often fail to expect the outcomes of their actions without directly executing them within the web environment. Recent approaches, such as the Tree Search Agent, can resolve this issue by exploring future states through environment interaction but often rely on trial-and-error, which can lead to undesirable outcomes. For instance, an agent tasked with booking a flight from Illinois to Vancouver might mistakenly purchase a non-refundable ticket for the wrong date. To allow web agents to avoid such unexpected outcomes, in our recent work [4], we proposed web agents integrated with world models which simulate the state transition of the action candidates for better decision-making. A promising direction that extends this idea is to train a world model that predicts the environment through a format of video, similar to Sora from OpenAI. By simulating the web environment entirely, such models could eliminate privacy risks associated with accessing individual computers while providing a robust framework for decision-making.

Universal Reward Models for Digital Agents. One possible next step is to build a universal policy agent for computer tasks by training LLMs on massive expert trajectories (i.e., behavior cloning). However, there is no sufficiently large dataset of expert trajectories. In addition, there is no guarantee that the ability to achieve a long-term goal is learned only with supervised fine-tuning [2, 6]. Thus, I am currently working on collecting a massive pairwise preference dataset for web agents and building foundational reward models for digital agents from it. I believe the reward model trained on extremely diverse websites and instructions will generalize well to unseen websites and tasks. With this reward model, we can not only select the best action (e.g., Best-of-N sampling) from the existing policy model but also train a more reliable policy model using RL with our foundational reward model. In the future, I aspire to develop universal reward models capable of tackling all tasks within a computer environment, extending beyond web navigation.

3 Future Plans at Georgia Tech

Why Georgia Tech? My curiosity-driven passion for exploring code-assisted digital agents, combined with my experience in the field, makes the PhD program at Georgia Tach the ideal environment to fuel my ambition of pushing boundaries and tackling intellectually challenging problems with outstanding faculties of similar interests. My work on building reward models for the agents is primarily based on Professor X, who is one of the faculties I would like to collaborate with. Also, I am looking forward to working with Professor X, as his extensive knowledge and insights on data-efficient learning and safety in interactive systems. Furthermore, Professor X's groundbreaking work in controllable generation has profoundly influenced and guided my previous research which aimed to have more control over LLMs' reasoning behavior.

Career Goals My long-term career goal is to become a professor, a role that perfectly aligns with my passion for collaboration, knowledge-sharing, and addressing critical issues through research. I envision a career where I can contribute significantly to the field of AI, mentor aspiring students, and foster an environment of innovation and discovery. My experiences and learning at Georgia Tech would be instrumental in shaping me into a well-rounded academic leader, equipped with the necessary skills and insights to make meaningful contributions to both academic community and the wider world.

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